Barely visible needles
Microneedle syringe and patch applications assured to enable pain-free injections and blood draws. **https://dgit.in/Dec20-15**

Scavenger hunt
Using AI-improved thermal imaging in drone cameras, to help find people lost in the woods. **https://dgit.in/dec20-16**

FEMTOSECOND CAMERAS

Researchers at Caltech have developed the fastest camera in the world

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For studying fundamental processes in physics, biology and chemistry, there is a requirement for really fast imaging techniques that can capture phenomena that occur over extremely short timescales. For example, femtochemistry deals with the chemical processes that occur over time scales that are one millionth of one billionth of a second, or one quadrillionth of a second. Essentially these ultrafast research cameras are on a quest to preserve time. The conventional approaches for the optical instruments use a number of pump-probe approaches. What this means is that it uses one laser to excite a target for observation, called the pump and another laser to investigate the consequences of that excitation, called the probe. Together, these two lasers provide the imaging information. Pump-probe technologies are used to investigate and characterise the properties of nanomaterials, semiconductors, biomolecules and pigments in paintings. However, the pump-probe approach is not suitable for capturing a wide range of physical phenomena including nuclear reactions, interactions of energetic particles, radioactive decay, nuclear explosions, rogue light waves, or light scattering in biological tissue.

Compressed ultrafast photography is a technique that is suitable for all these purposes that conventional pump-probe approaches do not work on.

Lihong Wang, a professor at Caltech had experience in biomedical imaging and developing systems that used a combination of light and sound, called photoacoustic imaging to capture structures such as tumours within the body. Around two years ago, Wang developed a new camera using a century old technique that resulted in a device capable of capturing one trillion frames per second. The underlying technique, called phase contrast microscopy was developed by the Dutch scientist Frits Zernike, who won the Nobel Prize in 1953 for the invention. Phase contrast microscopy